Chapter 3

Research Gap

The two foundations answer different questions. The biological foundation describes the circuit the model follows: a laminar column whose inhibition is divided among interneuron classes that act on different compartments of the pyramidal cell. The computational foundation describes what existing models do with sparse coding, interneuron classes and associative memory, and places them on three axes: substrate, inhibition model, and the arbitration between encoding and retrieval. This chapter draws the two together. It first sets out what is already established and is therefore not claimed here (Section 3.1), then shows where these strands fall short and states the gap (Section 3.2), and finally answers the objections that a reader familiar with the field is most likely to raise (Section 3.3).

3.1 What Is Established

Most of the components of this thesis are not new, and the claim made here depends on keeping them apart from what is.

The compartmental division of inhibition is established biology. PV basket and chandelier cells inhibit the perisomatic region and the axon initial segment, SST Martinotti cells inhibit the apical tuft, and VIP cells disinhibit the tuft by inhibiting SST cells [1] (Section 1.3). That PV and SST cells play different roles in a circuit is also established in models. Rate models with multi-compartment pyramidal cells have shown how the VIP, SST and PV motif gates pathways and redistributes inhibition between soma and dendrite [2, 3], and laminar spiking models now place PV, SST and VIP populations in every layer of a column [4, 5].

The two operators that the thesis assigns to these classes are established as well. Divisive normalisation is a canonical neural computation [6], and its link to PV cells has direct, if contested, experimental support [7]. Dendritic gating of top-down input at the apical tuft is established at the level of single cells and circuits [8, 9, 10], and has been turned into a learning principle in

which SST-like interneurons act on the apical compartment [11, 12].

The use of interneuron classes inside a memory network is not new either. In the attractor and BCPNN tradition, basket cells implement competition between minicolumns, Martinotti cells set how long an attractor lasts, and double bouquet cells make learned inhibition compatible with Dale's principle [13, 14]. Finally, the substrate is taken over unchanged from the HTM tradition: sparse distributed representations over minicolumns with dendritic segments [15, 16, 17], and the separation of encoding and retrieval as a functional requirement of memory has a clear precedent in the hippocampal model of Hasselmo et al. [18]. Cutsuridis et al. [19] have shown, in a model of hippocampal CA1, how PV-type and SST-type interneurons can divide the work of the two modes between them.

3.2 Where the Strands Fall Short

Each of these strands, then, is well developed on its own terms. What they have not done is come together. Table 3.1 shows this by reducing Table 2.1 to a single row per tradition, set against the three parts of the gap. No tradition covers more than two of them. The one model that does test how interneuron classes share encoding and retrieval is a biophysical model of hippocampal CA1, not a neocortical SDR column.

Put together, the gap has three parts: (i) a discrete SDR substrate in the HTM tradition; (ii) inhibition split into two interneuron types with distinct operations, a PV-like divisive normalisation on the output compartment and an SST-like dendritic gate on the apical input compartment; and (iii) an explicit test of whether these two operations contribute differently to encoding and to retrieval. This thesis sets out to hold all three at once. In practice, that means taking a biological distinction that is already well established, carrying it into a substrate that has so far reduced inhibition to a single sorting rule, and asking whether the move changes anything that homogeneous kWTA could not. As in the hippocampal models, which mode is active is set from outside the column, by the encoding or retrieval condition presented to it. What the thesis tests is how the two interneuron types share the work within each mode, and, where feedforward and recalled signals arrive together, how the column integrates them.

The contribution is therefore a modest one. The thesis does not offer a new account of cortical inhibition or a new memory model, but a controlled substitution inside an existing one, and the outcome is useful either way. If the two operations do divide encoding and retrieval between them, the SDR substrate gains a cell-type-specific division of labour between encoding and retrieval, of the kind Cutsuridis et al. describe for CA1. If they do not, that is evidence that, at least in this setting, the homogeneous rule loses less than the biology would suggest.

Tradition	**(i) Discrete SDR substrate**	**(ii) Operator-distinct PV and SST**	**(iii) Differential PV/SST roles in encoding and retrieval tested**
Hippocampal CA1 model	No: biophysical spiking cells	Partly: PV-type and SST-type cells, not formal operators	Yes, with the mode set by septal theta
HTM, CLA and SDR models	Yes	No: homogeneous kWTA	No: single inhibition rule
SDM and NAA	Yes (NAA); abstract binary (SDM)	No	No: single inhibition rule
Interneuron-resolved circuit models	No: rate or spiking populations	Yes	No: no memory task
Associative-memory columns	Partly: minicolumns, no learned SDRs	Partly: several classes, roles in retrieval dynamics	No: storage and recall roles not compared
Operator mechanisms	No	Partly: operators defined, cell attribution contested	No: no memory task
This thesis	**Yes**	**Yes**	**Tested, with the mode set externally**

Table 3.1: The research gap at a glance. Each row condenses one tradition from Table 2.1; the columns are the three parts of the gap stated in the text.

3.3 Anticipated Objections

This has already been done in the hippocampus. Cutsuridis et al. [19] assign storage and recall to different interneuron classes in CA1 and test those roles by removing the classes one at a time. This is the closest precedent for the thesis, and the thesis follows it in letting the mode be set from outside the circuit. The model, however, is a compartmental spiking model of a hippocampal region whose inputs arrive in separate dendritic layers from entorhinal cortex and CA3, and its interneurons are detailed cell models rather than operators that could replace a selection rule. The question of what happens when the single inhibition rule of an SDR column is replaced by two such operators, in a neocortical laminar setting, does not arise there.

PV and SST have already been dissociated. Hertäg and Sprekeler [3] model PV cells on the soma and SST cells on the apical dendrite, and show that the circuit can switch persistently between integrating and cancelling top-down input. This is the closest precedent for a circuit-internal switch and the thesis treats it as such. Its substrate, however, is a rate model with no sparse distributed representations and no learned associations, and the switch concerns how feedback is processed, not whether a pattern is stored or recalled.

Interneuron classes have already been studied in memory networks. Krishnamurthy et al. [13, 20] show that basket and Martinotti cells shape attractor memory differently, with Martinotti cells setting dwell time and producing surround suppression. Here the memory setting is real, but the roles concern the dynamics of a pattern that is already being recalled, in a spiking attractor network whose patterns were not learned as SDRs, and the change of mode in this tradition comes from neuromodulation [21].

The mechanisms are borrowed rather than discovered. This is correct and intended. The thesis does not propose a new form of inhibition. Its claim concerns the consequence of placing two known operators in a substrate that lacks them, which is a question the source models cannot answer because they do not use that substrate. A negative result, in which the two operators behave no differently from kWTA with respect to encoding and retrieval, would be as informative as a positive one.

PV is not purely divisive and SST is not purely subtractive. The arithmetic of each class depends on stimulus and network state [22, 23], and the attribution of division to PV cells is contested [24, 25]. The thesis therefore makes its claim about operators, not about cell classes. The operators

are called PV-like and SST-like because their targets follow the compartmental targeting described in Section 1.3, which is robust, while the mapping of each operator to a single class is a modelling assumption on which the results about the operators do not depend.

The HTM tradition already names its inhibitory cells. Hawkins and Ahmad [16] identify fast-spiking basket cells as the likely source of the inhibition their model needs, and the author of NAA leaves the inhibition model explicitly open [26]. Naming a cell type is not implementing it: in both cases the computation remains a homogeneous k-winners-take-all rule (Section 2.2.5).

The aim of the thesis follows directly from this gap and is stated in the following chapter.

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